A Totally Inconspicuous Bag FAMMAE2250ODPFPTT1

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Context: Why Spotted Lanternflies are a Problem

Spotted lanternflies (SLF), are an invasive pest damaging grape farms along the U.S. East Coast. Their instars and adults feed on grapevine sap, while honeydew from late-stage SLF causes sooty mold making crops unsellable. A single SLF in a 22-ton grape harvest can cause rejection, and untreated vines may host about 22.9 SLF each.[1] Mechanical separation of SLF from harvests is ineffective due to their small size (~1 inch) and does not prevent crop damage. Insecticides are costly, ranging from $16.13 to $379.97 per acre every year, and often fail to fully control infestations. To put this into perspective, a 100-acre vineyard which is treated for the year could result in costs ranging from $1,600 - $38,000 per growing season just on the insecticide. Even a single rejected 22-ton harvest could result in tens of thousands of dollars in lost revenue and profit.

Problem Statement: Prevention over Reaction

Our solution aims to prevent SLF from reaching grapevines, preventing vineyard deterioration and the inconveniences associated with harvest contamination. Our design consists of a plastic bag containing methyl salicylate (wintergreen oil) attached to a brace and funnel with a speaker emitting 60 Hz [Figure 1]. SLF are attracted to wintergreen oil as it acts as a kairomone, a chemical signal which mimics cues from their preferred host plants.[2] 60 Hz sound waves are also found to attract SLF.[3] Our design will be hung onto a tree using a rope and hook attachment [Figure 2]. Our solution targets the 4th instar, which is the stage in the SLF life cycle in which the SLF migrated to woody plants.[4] We focus on this specific lifecycle stage of the SLF because during this stage, the SLFs tend to prioritize and migrate towards woody plants, funneling them into a target we can isolate and localize. Regardless however, our solution is able to target all stages of the SLF life cycle. Our solution's main constraints were cost, user interaction, and ease of use. We wanted a low-cost solution farmers would be able to easily install across their farm to capture SLFs.

Figure 2. Rope and pillar attachment

Figure 1. Final Design: A Totally Inconspicuous Bag

Final Prototype, Application, and Assembly Process

SLF in their instar stage frequently fall out of their host trees. Our funnel will hang near the base of the tree and attract these SLF whilst also ensuring, once they enter, they can't escape through the same opening. Trap bags are disposable and can easily be removed and replaced without removing the brace from the tree whilst ensuring the SLF don't escape [Figure 3].

To assemble the product, simply slide the speaker and battery module into the carriage. Slot the funnel and speaker carriage into the acrylic back panel, such that the funnel and speaker carriage are

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[1] Bekelja, K., & Russo, J. P. (2026, January). NY Grapes, and the Spotted Lanternfly Problem. MAE2250. Ithaca; Hollister Hall.

[2] Burga, S. (2024, July 1). Scientists are still trying to figure out how to kill invasive spotted lanternflies. https://time.com/6994293/spotted-lanternfly-scientists-research-kill/

[3] Elliott, S. (2024, January 19). Spotted lanternfly reveals a potential weakness. https://www.usda.gov/about-usda/news/blog/spotted-lanternfly-reveals-potential-weakness

[4] "Spotted Lanternfly Lifecycle." Rainbow Ecoscience,www.rainbowecoscience.com/education-and-training/spotted-lanternfly/spotted-lanternfly-lifecycle?srslti=AfmBOoobEB2WTLMtLEBQEmkjqt2KFI4xGGjYdsp8VSobur-IsZPa4fxM. Accessed 4 May 2026.

facing the same direction. Tie a rope around the tree or object that the product is being hung on. Place a bag with the open end around the funnel and slide the closed end behind the speaker carriage. If needed, such as when operating in rainy or windy conditions, zip tie the end of the bag onto the ledge built into the funnel. As this design relies on attracting SLFs, several will need to be placed around the perimeter of a target area around 10m apart so that SLFs are lured into our traps rather than areas of interest, such as grapevines.

Conclusion and Recommendations

This design does effectively meet our goal of attracting SLFs whilst being more affordable than conventional insecticides. The simplicity of our design and materials [refer to Bill of Materials for costs] makes this more affordable and viable for small farmers as well. However, there are still things to improve on the design.

The speaker and circuit makes up the majority of the cost of the design (~50% of the cost). This is largely due to the variety of parts we had to separately order. We can likely simplify the speaker circuit if we had dedicated manufacturing lines specifically for 60 Hz speakers/buzzers. With this, we can lower the cost even further as well as ensure the reliability of our speakers to produce a 60 Hz vibration.

Battery life is also an issue as the speakers are running all day and batteries can be costly to replace. Future prototypes can use solar powered batteries, ideal for the tall structures the traps are meant to be placed on. Solar powered batteries would also line up with their most active times being from noon to dusk.[5] This would remove more human involvement and make this much more convenient to use, only having to replace the bag once in a while.

Placement of the trap is also a little difficult especially for a single individual when the tree or tall object is large. It is also somewhat unreliable as the trap may slip down if the user doesn't tie a knot properly. Future prototypes will get rid of the hook and replace it with a sticky pad on the back making placement easier. As the only thing needing replacement is the bag, the design doesn't need to be removed and so the sticky pad should still remain sticky even under the elements.

Testing and Results

We conducted three key tests to evaluate the effectiveness of our prototype: the speaker's performance, structural load test, and a water resistance test:

Our first, evaluated our speaker's ability to produce 60 Hz frequency with ±5 Hz range, the same shown to be capable of attracting spotted lanternflies. Our results were inconclusive, with the speaker likely producing the correct frequency, but being very quiet and producing lots of extra unneeded frequencies. Thus, our recommendation would be to incorporate a higher quality speaker and array for producing a pure 60Hz sine wave.

Our second test evaluated our design's ability to withstand structural load. We tested its ability to withstand load up to 1 kg without structural failure, roughly the weight of a full bag of spotted lanternflies. This test was successful and there do not seem to be any weaknesses in the design's ability to withstand structural loading.

The final test was for the elements, the bag must be able to withstand stormy conditions and still not trap water in the bag, nor allow any water to reach the speaker. We conducted this by holding the bag under running water for 30 seconds. This test was also successful as no water made its way into the bag, nor was able to reach the speaker.

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[5] Entomology. (2026, March 9). *Spotted lanternfly*. Spotted Lanternfly. https://www.in.gov/dnr/entomology/pests-of-concern/spotted-lanternfly/

Initial Prototypes and Testing Details

Our first prototype was a mock-up to determine the general shape and characteristics of our design [Figure 3]. The design included a hook; speaker housing, brace, and funnel made of cardboard, and a gallon ziploc trap bag. We determined that the speaker housing needed to be made larger and the trap bag needed to be attached more rigidly to the design.
We utilized a 3D printer to create the speaker housing, brace, and funnel for the first functional prototype [Figure 6]. The bag was still not rigidly connected due to the funnel's smooth surface and the speaker housing was not large enough to accommodate the required batteries. Per our testing, the hook attachment could not accommodate 1kg of weight. We made necessary changes, outlined in the final prototyping section above, to create our final trap.

Figure 3. First Prototype

Figure 6. Second Prototype

Test Details:

Our target frequency for the speaker performance test was 55-65Hz; the range most effective in attracting spotted lanternflies. Our first functional prototype contained a bluetooth speaker, only able to consistently produce a signal of about 80Hz[6] [Figure 4]. To test our final prototype's frequency we used a test tone outputted from a tone generation software[7], tested against our final prototype. However, due to poor microphone quality and signal clipping the tone showed as an impure 180Hz frequency on our analysis software[8][Figure 5]. This is due to signal clipping which triplicates the frequency of the signal, and continues to show waves at its harmonics, validating the production of a 60 Hz frequency. However, due to the difficulty to produce a loud tone and lack of quality, it would still be our recommendation to improve the speaker and frequency producing system.

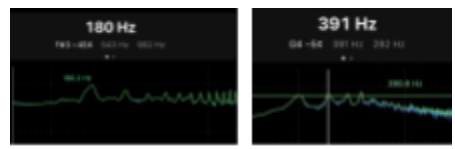

Figure 5. Test tone (left) and final (right) frequency analysis

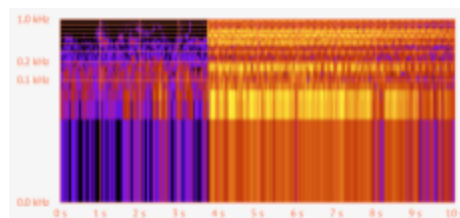

Figure 4. P2 Frequency Analysis

We tested whether our trap's brace could withstand a load of 1kg, the estimated weight of a gallon bag full of spotted lanternflies, without experiencing static failure by hanging the prototype on a pillar and weighing it down with 1 liter of water. The results of this test showed that the design was more than strong enough to withstand the loads, and that our prototype has a factor of safety allowing it to carry the weight of the lanternflies as well as additional loads from the weather (high winds).

To confirm that the trap is water resistant, we tested whether water would penetrate the speaker housing or collection bag after 60 seconds under running water. The final prototype successfully prevented water intrusion, demonstrating that our drainage system can withstand environmental conditions. Based on the findings, our current design is effective and shows promise for future development. Compared to the high recurring costs of insecticide solutions, further refinement of this prototype will remain cost-effective and a potential solution for SLF mitigation.

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[6] *Free online audio spectrum analyzer*. MAZTR. (n.d.).https://www.maztr.com/audiospectrumanalyzer#google_vignette
[7] Szynalski, T. P. (n.d.). Online tone generator. Online Tone Generator - generate pure tones of any frequency. https://www.szynalski.com/tone-generator/
[8] Manfred Kuegel. (2021). Frequency Analyzer [Mobile app]. Apple App Store. https://play.google.com/store/apps/details?id=com.lexi.android.

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Bill of Materials

Per Unit Cost

Component	Per Unit Cost ($)	Quantity	Total Cost ($)
9 Volt Battery	12.69 for 8	1/4	3.17
3D Printed Funnel	N/A	1	4.73
3D Printed Speaker Carriage	N/A	1	10.87
Speaker	15.00 for 4	1/4	3.75
Wire	0.30 per foot	1/2	0.15
Pulse Generator	10.97 for 2	1/2	5.485
Rope	12.50 for 50 ft	1	12.50
Bag	6.95 for 50	1/50	0.14
Total			40.80

Total including R&D

Amazon BOM

Description	Justification	Quantity	Unit of measurement	Total Cost
Pulse Generator	To attract SLF to our trap - None found on McMaster.	1	2 Pulse Generators	$10.97
Bluetooth Speaker	To attract SLF to our trap - None found on McMaster that are to the scale of our project	1	Each	$6.99
Zipties	To attach trap bag to speaker frame - McMaster significantly exceeds our budget	1	100 zip ties	$3.99
Duct Tape	To secure frame - McMaster significantly exceeds our budget	1	3 Rolls of 1.88" x 30-yard	$13.36
Methyl Salicylate 99% Pure Premium Wintergreen Oil	To attract SLF to our trap - None found on McMaster.	1	100 mL / 1 Bottle	$15.00
BOJACK NE555 Timer IC NE555P Pulse	Pulse generator to attract SLF - None found on McMaster.	1	8 Pack	$6.99
24 Values Electrolytic Capacitor Assortment Kit	For speaker - McMaster significantly exceeds our budget	1	Each	$9.99
WWZMDiB 80 Pcs Potentiometer Kit with Knobs	For speaker - McMaster significantly exceeds our budget	1	Each	$8.99
4 Speakers V1	To attract SLF to our trap - Those found on McMaster exceeded the size scale of our project	1	4 Speaker	$9.99
Electronic components kit	For speaker - McMaster significantly exceeds our budget	1	Each	$15.99
9V Batteries	For speaker - McMaster significantly exceeds our budget	1	8 Pack	$12.69

General Purpose Transistor	For speaker - McMaster significantly exceeds our budget	1	Pack of 200 each	$6.99
4 Speakers V2	To attract SLF to our trap - Those found on McMaster exceeded the size scale of our project	1	4 Speakers	$15.99
Mini MP3 Player	To attract SLF - None found on McMaster	1	Each	$9.90
Amplifier Board	To attract SLF - None found on McMaster	1	2 Boards	$12.99
USB Receiver	For speaker - McMaster significantly exceeds our budget	1	3 Pieces	$5.99

McMaster BOM

Description	McMaster Code	Quantity	Unit of measurement	Total Cost
UV-Resistant Sure-Grip Rope | To attach our trap to trees	3818T23	1	50 ft	$12.50
Resealable Bags | To hold SLF that we trap	1959T63	1	Pack of 50	$6.95

3D Printed BOM

Component	Quantity	Total Cost ($)
3D Printed Funnel	2	9.46
3D Printed Speaker Carriage	2	21.74
Total		31.20

<u>Total Cost for full development of our final product: $217.46</u>